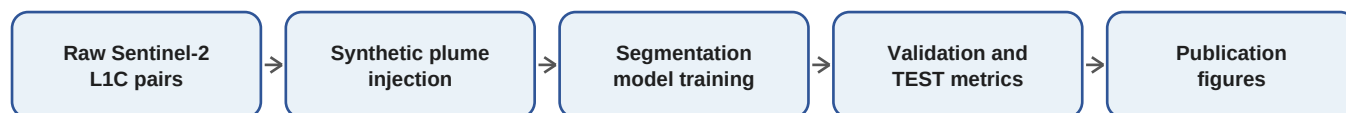


Beginner Guide to the CH4 Plume Synthetic Publication Project

A step-by-step explanation of the new synthetic methane plume segmentation project, what each technical term means, why this route is different from the older projects, and how the full pipeline works.



Main idea: this project uses real satellite images as backgrounds, injects artificial methane plumes in a controlled way, and trains models to find those injected plumes.

1. The Project In One Sentence

The new project, named CH4_Plume_Synthetic_Publication, is a self-contained benchmark for methane plume segmentation using synthetic plumes injected into real Sentinel-2 L1C event/reference image pairs.

This means the images are real satellite backgrounds, but the plume labels are not transferred from Carbon Mapper or EMIT. Instead, the project creates its own plume signal and its own perfectly aligned plume mask.

2. Beginner Definitions

Term	Beginner meaning	Why it matters
CH4	Chemical formula for methane.	Methane is the gas we are trying to detect as a plume.
Plume	A cloud or trail of gas spreading away from a source.	The model must identify the plume region in the image.
Segmentation	Assigning a label to every pixel in an image.	The output is a map saying plume or not plume for each pixel.
Synthetic	Artificially generated, not directly observed.	The plume is created by code so the label is perfectly known.
Sentinel-2	A European Earth-observation satellite mission.	It provides the optical/SWIR imagery used as background data.
L1C	Level-1C top-of-atmosphere Sentinel-2 data.	It is close to the measured satellite signal before surface correction.
SWIR	Short-wave infrared light.	Methane has absorption behavior in SWIR wavelengths.
B11/B12	Sentinel-2 SWIR bands.	These bands are commonly used in Sentinel-2 methane-sensitive features.
Event image	The satellite image near the plume event date.	This is where the synthetic plume is injected.
Reference image	A comparable image without the target event.	It helps remove background surface patterns.
GeoTIFF	A geospatial image file with coordinate information.	The raw image chips are stored as .tif files.
Manifest	A CSV table listing data files and metadata.	The code uses it to know which files belong to TRAIN, VAL, and TEST.
TRAIN	Data used to teach the model.	The model adjusts its weights using this data.

VAL	Validation data used during training.	It helps choose the best checkpoint.
TEST	Held-out data used after training.	It gives the final benchmark result.
Tau	Optical-depth strength of the synthetic methane plume.	Higher tau means stronger simulated absorption.
Benchmark	A fair comparison between models.	PhysTAUNet is compared against U-Net, Attention U-Net, U-Net++, and DeepLabV3+.

3. Why A New Synthetic Project Was Needed

Earlier work tried to use real plume information from Carbon Mapper and EMIT as labels for Sentinel-2 segmentation. The problem is that those real labels are not automatically pixel-perfect Sentinel-2 labels.

- Carbon Mapper and EMIT are different sensors from Sentinel-2.
- They may observe the plume at a different time.
- They may have different pixel sizes and viewing geometry.
- Their plume masks may shift after reprojection onto the Sentinel-2 grid.
- Methane plumes can move quickly with wind, so a label from one time may not match a Sentinel-2 image from another time.
- For segmentation, even small shifts can make precision, recall, and F1 look bad.

The synthetic project avoids that label-alignment problem. It creates the plume directly on the Sentinel-2 image grid, so the model target is known exactly.

4. How This Project Is Different From The Older Projects

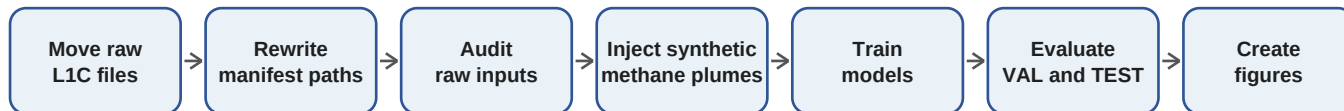
Project	Main goal	Input	Output	Main limitation
CH4_Plume_Publication	Methane enhancement regression.	Sentinel-2 plus TROPOMI-style data.	Continuous CH4 enhancement in ppb.	Not true plume segmentation.
CH4_Plume_Segmentation_Publication	Real plume-mask segmentation using Carbon Mapper and EMIT labels.	Sentinel-2 images and real external plume labels.	Plume mask.	Cross-sensor labels were not reliable enough at pixel level.
CH4_Plume_L1C_Publication	Use Sentinel-2 L1C event/reference pairs for plume segmentation.	L1C event/reference data and real labels.	Plume mask.	Still affected by real-label transfer issues.
CH4_Plume_Synthetic_Publication	Controlled synthetic plume segmentation benchmark.	Real L1C backgrounds plus injected synthetic plumes.	Perfectly aligned synthetic plume mask.	Claim must stay synthetic, not operational real detection.

5. What Self-Contained Means Here

Self-contained means that after the import step, the new project should not depend on the old project for training. The raw data files are moved into the new project, and the manifests are rewritten so all file paths point inside CH4_Plume_Synthetic_Publication.

```
CH4_Plume_Synthetic_Publication/  
data/raw/sentinel2_l1c/exports/ raw L1C GeoTIFF image chips  
data/training_l1c/ rewritten manifests and audits  
data/models/ trained model checkpoints  
data/outputs/ tables and publication figures  
reports/ markdown and PDF reports  
src/ch4l1c/train.py model and synthetic plume code  
scripts/run_all.sh full pipeline entrypoint
```

6. Step-By-Step Pipeline



Step 1 - Move raw data

The raw Sentinel-2 L1C GeoTIFF files are moved from CH4_Plume_L1C_Publication into CH4_Plume_Synthetic_Publication. Moving avoids filling the disk with duplicate copies.

Step 2 - Rewrite manifests

The CSV manifests are copied and edited so each s2_path points to data/raw/sentinel2_l1c/exports inside the synthetic project.

Step 3 - Audit inputs

The audit checks that every manifest row points to a real file and that paths are inside the new project.

Step 4 - Generate synthetic plume

For every training patch, the code creates an artificial plume field with irregular shape.

Step 5 - Inject methane absorption

The code modifies the methane-sensitive SWIR signal to simulate the effect of methane absorption.

Step 6 - Train models

PhysTAUNet and benchmark models learn to predict the synthetic plume mask.

Step 7 - Validate and test

Validation tracks training quality. TEST gives held-out final benchmark metrics.

Step 8 - Create outputs

The pipeline writes CSV tables, reports, metric plots, and qualitative example figures.

7. The Synthetic Plume Method

The method is physics-inspired. It does not pretend that the synthetic plume is an observed real plume. Instead, it uses a simplified methane absorption model to create a controlled learning task.

The most important formula in the current code is the attenuation step:

```
absorption = exp(-tau * plume_field)
injected_B12 = event_B12 * absorption
```

Beginner meaning: the synthetic plume reduces the methane-sensitive B12 signal. The reduction is stronger where the plume field is stronger and where tau is larger.

- `plume_field` is the spatial shape of the plume, with higher values near stronger plume regions.
- `tau` controls the strength of methane absorption.
- `exp` means exponential function. It makes absorption smoothly decrease as `tau` and plume strength increase.
- `event_B12` is the original Sentinel-2 B12 band in the event image.
- `injected_B12` is the modified B12 band after the synthetic methane plume is added.

8. Why The Shape Generator Was Changed

Earlier synthetic examples looked too regular, almost like smooth ellipses. That is not realistic enough because real methane plumes are shaped by wind, turbulence, source behavior, and background noise.

The current generator creates more realistic variation:

- meandering centerline - the plume path curves instead of staying straight
- source/core region - a stronger area close to the emission source
- width variation - the plume can widen or narrow downwind
- detached wisps - small separated patches can appear away from the main plume
- turbulence texture - the intensity is not perfectly smooth
- holes and ragged edges - boundaries are irregular instead of clean ellipses
- random thresholding - each mask can have a different final outline

9. What The Model Actually Learns

The model is not learning from real Carbon Mapper plume masks in this project. It learns a controlled task: given a real Sentinel-2 background with synthetic methane absorption injected, predict the synthetic plume mask.

Input to model:
real Sentinel-2 L1C event/reference features
plus synthetic methane absorption signal

Target label:
synthetic plume mask created from the same plume field

Model output:
probability map, where each pixel receives a plume probability

A probability map is not immediately a final mask. A threshold converts probabilities into plume/non-plume pixels. For example, if threshold = 0.5, then pixels with probability ≥ 0.5 are predicted as plume.

10. Models In The Benchmark

Model	Beginner explanation
U-Net	A common medical and remote-sensing segmentation model. It uses an encoder-decoder shape with skip connections.
Attention U-Net	A U-Net variant that adds attention gates, helping the model focus on relevant regions.
U-Net++	A U-Net variant with more nested skip pathways, designed to improve feature reuse.
DeepLabV3+	A segmentation model that uses atrous/dilated convolutions to capture broader context.
PhysTAUNet	The custom model. It is meant to use methane-sensitive physics features and segmentation learning together.

11. Metrics Explained

Metric	Meaning	Beginner interpretation
Precision	Of all pixels predicted as plume, how many were truly plume?	High precision means few false plume pixels.
Recall	Of all true plume pixels, how many did the model find?	High recall means the model misses fewer plume pixels.
F1	A balance between precision and recall.	Useful when plume pixels are rare.
IoU	Intersection over Union between predicted and true plume masks.	A stricter spatial-overlap score.
Tolerant F1	F1 after allowing small spatial mismatch.	Useful because plume boundaries can be uncertain.
Predicted positive fraction	Fraction of pixels predicted as plume.	If this is near 1.0, the model is predicting almost everything as plume.

12. Why The Smoke Test Looked Bad

The smoke test used only 20 raw files, 1 epoch, and 64 patches per epoch. That run was only meant to check whether the pipeline works. It was not meant to produce a scientific result.

In that smoke run, `predicted_positive_fraction` was close to 1.0. That means the model learned a trivial behavior: predict nearly every pixel as plume. This is a common early-training problem when the run is too short.

The real benchmark needs many more patches and more epochs, for example 40 epochs and 8192 patches per epoch.

13. Commands To Reproduce

Install and activate environment:

```
cd ~/CH4_Plume_Synthetic_Publication
source /home/ubuntu/.venv/bin/activate
python -m pip install -r requirements.txt
python -m pip install -e .
```

Full self-contained run, moving raw files instead of duplicating them:

```
cd ~/CH4_Plume_Synthetic_Publication
source /home/ubuntu/.venv/bin/activate

SOURCE_PROJECT=~ /CH4_Plume_L1C_Publication \
IMPORT_MOVE_RAW=1 \
IMPORT_OVERWRITE=1 \
EPOCHS=40 \
BATCH_SIZE=8 \
PATCH_SIZE=128 \
PATCHES_PER_EPOCH=8192 \
VALIDATION_PATCHES=1024 \
TEST_PATCHES=2048 \
MODELS="phys_tau_net unet attn_unet unet_pp deeplabv3p" \
bash scripts/run_all.sh
```

Check outputs after training:

```
cat data/outputs/tables/synthetic_publication_benchmark_summary.csv
ls -lh data/outputs/publication_figures/
ls -lh data/outputs/publication_figures/synthetic_segmentation_examples/
```

14. Outputs Explained

Output	What it contains
data/models/synthetic_publication_benchmark/{model}/best.pt	Best saved model checkpoint for that architecture.
data/models/synthetic_publication_benchmark/{model}/history.csv	Validation metrics after each epoch.
data/models/synthetic_publication_benchmark/{model}/synthetic_eval_test.csv	Held-out synthetic TEST metrics.
data/outputs/tables/synthetic_publication_benchmark_summary.csv	Combined benchmark table across all models.
data/outputs/publication_figures/metrics_synthetic_segmentation.png	Validation metric figure.
data/outputs/publication_figures/metrics_synthetic_test_segmentation.png	Held-out TEST metric figure.
data/outputs/publication_figures/synthetic_segmentation_examples/	Qualitative segmentation examples for publication review.
reports/synthetic_publication_benchmark_report.md	Markdown report summarizing the benchmark.

15. What Can Be Claimed In A Paper

A safe and accurate publication claim is:

PhysTAUNet was benchmarked against U-Net, Attention U-Net, U-Net++, and DeepLabV3+ on a controlled synthetic Sentinel-2 L1C methane plume segmentation task. Synthetic plumes were injected into real Sentinel-2 L1C event/reference backgrounds using methane-sensitive SWIR attenuation and irregular plume morphology.

Do not claim that this project proves operational real-world plume segmentation from Carbon Mapper or EMIT labels. That would require a separately validated real-label dataset.

16. Limitations

- Synthetic plumes are controlled and useful for benchmarking, but they are not the same as observed real plumes.
- The injection model simplifies methane radiative transfer. It is physics-inspired, not a full atmospheric radiative-transfer simulator.
- Model performance on synthetic TEST data does not automatically prove performance on real methane events.
- Future real-world validation still needs carefully matched, pixel-reliable plume labels.
- The paper should be clear that the main claim is synthetic Sentinel-2 plume segmentation.

17. Recommended Next Work

- Improve the synthetic generator using wind direction, source location, and physically parameterized plume spread.
- Compare synthetic plume shapes against real Carbon Mapper and EMIT morphology statistics.
- Add ablation studies: with/without event-reference features, with/without tau physics features, and with different plume strengths.
- Create a small manually reviewed real Sentinel-2 validation set if possible.
- Use the synthetic benchmark to justify PhysTAUNet architecture, then treat real-data testing as a separate follow-up study.

18. References For Method Framing

- Climate Change AI / NeurIPS 2023: Methane plume detection with U-Net segmentation on Sentinel-2. <https://www.climatechange.ai/papers/neurips2023/78>
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- Wang et al. 2024: Matched filter for Sentinel-2 methane plume detection. <https://www.mdpi.com/2072-4292/16/6/1023>
- Sentinel-2 methane monitoring physics and B11/B12 transmittance context. <https://amt.copernicus.org/articles/16/89/2023/>